



Experiment -1

Student Name: Sahilpreet Singh

UID: 24BCS10476

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Subject Name: Salesforce-1

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1. **Aim:** To implement SQL DDL, DML, DQL, and DCL commands by creating and populating a Hospital Management database with integrity constraints, and to retrieve specific records using SQL queries to understand relational database operations.

2. **Requirements (Hardware/Software):** Software: CodeChef

3. Code:

--Complete the queries below to insert data into the tables & retrieving the first records from the first 3 tables.

-- Inserting Data into Doctors Table

```
INSERT INTO Doctors (DoctorID, Name, Specialization, ContactNumber, Email) VALUES  
(1, 'Dr. John Smith', 'Cardiology', '1234567890', 'john.smith@hospital.com'),  
(2, 'Dr. Lisa Brown', 'Neurology', '0987654321', 'lisa.brown@hospital.com');
```

-- Inserting Data into Patients Table

```
INSERT INTO Patients (PatientID, Name, DOB, Gender, ContactNumber, Address) VALUES  
(1, 'Alice Johnson', '1990-05-21', 'Female', '1112223333', '123 Main St'),  
(2, 'Bob Martin', '1985-08-14', 'Male', '4445556666', '456 Elm St');
```

-- Inserting Data into Appointments Table

```
INSERT INTO Appointments (AppointmentID, PatientID, DoctorID, AppointmentDate, Status)  
VALUES  
(1, 1, 1, '2025-02-15', 'Scheduled'),  
(2, 2, 2, '2025-02-16', 'Completed');
```

-- Inserting Data into Treatments Table

```
INSERT INTO Treatments (TreatmentID, PatientID, DoctorID, Diagnosis,
TreatmentDescription, TreatmentDate) VALUES
(1, 1, 1, 'Hypertension', 'Prescribed medication', '2025-02-15'),
(2, 2, 2, 'Migraine', 'MRI Scan and medications', '2025-02-16');
```

-- Inserting Data into MedicalRecords Table

```
INSERT INTO MedicalRecords (RecordID, PatientID, TreatmentID, Notes) VALUES
(1, 1, 1, 'Patient responding well to treatment'),
(2, 2, 2, 'Further evaluation required');
```

-- Inserting Data into Billing Table

```
INSERT INTO Billing (BillID, PatientID, TreatmentID, Amount, BillDate, Status) VALUES
(1, 1, 1, 200.00, '2025-02-15', 'Paid'),
(2, 2, 2, 500.00, '2025-02-16', 'Unpaid');
```

--Now write the queries for retrieving the first records from the first three tables (Doctors, Patients, Appointments).

```
SELECT * FROM Doctors WHERE DoctorID = 1;
```

```
SELECT * FROM Patients WHERE PatientID = 1;
```

```
SELECT * FROM Appointments WHERE AppointmentID = 1;
```

4.Output:

The screenshot shows a SQL IDE interface with a dark theme. On the left, the 'Statement' tab is active, displaying the SQL queries. The 'Submissions' tab shows the results of the queries. The 'Solution' tab shows the expected output tables. The 'AI Help' tab is also visible.

Step 2:

After inserting the data, also write the queries for retrieving the **first records** from the **first three tables** (Doctors, Patients, Appointments).

Doctors Table

DoctorID	Name	Specialization	ContactNumber	Email
1	Dr. John Smith	Cardiology	1234567890	john.smith@hospital.com
2	Dr. Lisa Brown	Neurology	987654321	lisa.brown@hospital.com

Patients Table

PatientID	Name	DOB	Gender	ContactNumber	Address
1	Alice Johnson	21-05-1990	Female	1112223333	123 Main St
2	Bob Martin	14-08-1985	Male	4445556666	456 Elm St

Appointments Table

AppointmentID	PatientID	DoctorID	AppointmentDate	Status
1	1	1	2025-02-15	Scheduled

SQL

```
29 INSERT INTO Billing (BillID, PatientID, TreatmentID, Amount, BillDate, Status) VALUES
30 (1, 1, 1, 200.00, '2025-02-15', 'Paid'),
31 (2, 2, 2, 500.00, '2025-02-16', 'Unpaid');
32
33 --Now write the queries for retrieving the first records from the first three tables
34 (Doctors, Patients, Appointments).
35 select * from Doctors
36 where DoctorID=1;
37 select * from Patients
38 where PatientID=1;
39 select * from Appointments
40 where AppointmentID=1;
```

Your Output

PatientID	Name	DOB	Gender	ContactNumber	Address
1	Alice Johnson	1990-05-21	Female	1112223333	123 Main St

AppointmentID	PatientID	DoctorID	AppointmentDate	Status
1	1	1	2025-02-15	Scheduled

✓ Hooray, you did it!

Submit Next

5.Learning Outcome:

- Create and manage database tables using SQL Data Definition Language (DDL) commands.
- Insert, update, and delete records using SQL Data Manipulation Language (DML) statements.
- Retrieve and filter data efficiently using SQL Data Query Language (DQL) SELECT queries with appropriate conditions.
- Enforce data integrity by implementing Primary Key and Foreign Key constraints.
- Design and implement a relational Hospital Management System database to understand and apply core database concepts.